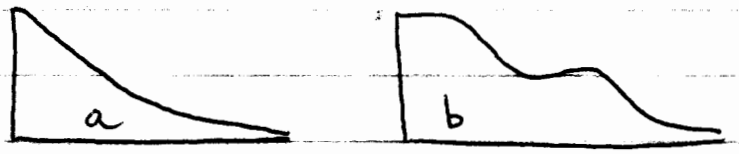
Solutions to HW12 ①

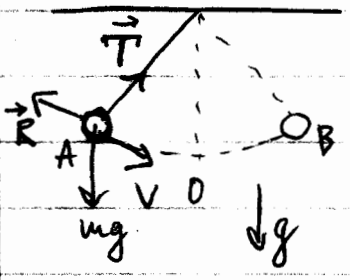
S&F, Ch. 5
Starykh

Q3



For a frictionless slide the final speed will be the same regardless of its precise shape because the only force doing work is gravitational, and the work done by it depends only on the height of the slide. With friction present, the longer slide will result in slower speed — this is because the friction does negative work (always!) and thus reduces kinetic energy.

Q7



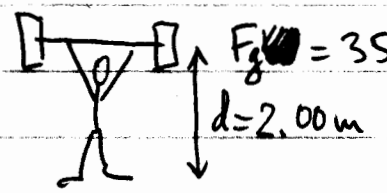
$\vec{T}$ = tension, $\vec{R}$ = air resistance force.

(a) $\vec{T}$ is perpendicular to the displacement of the suspended object ⇒ does no work

(b) $\vec{R}$ is opposite to the displacement (angle 180°, $\cos 180° = -1$) and hence does the negative work always

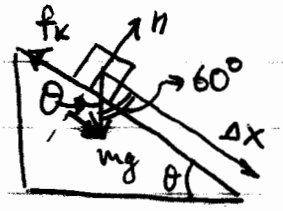
(c) $\vec{F}_g = m\vec{g}$ is along the displacement for the segment AO and does positive work there; along the OB one it is against displacement ⇒ does negative work.

P1



work $W = F_g \cdot d = 700$ J

P5



$m = 5.00\ kg$, $\Delta x = 2.50\ m$, $\theta = 30.0°$

$\mu_k = 0.436$

(a) $W_g = ?$ Component of mg along Δx is $mg \sin\theta$

$\Rightarrow W_g = mg \sin\theta \cdot \Delta x = 5.00\,kg \cdot 9.80\,m/s^2 \cdot \sin(30°) \cdot 2.50\,m =$
$= 61.25\ J$

Another way: the angle between mg and Δx is $90° - 30° = 60°$

$\Rightarrow W_g = mg \cos 60° \cdot \Delta x = 61.25\ J$ (because $\cos 60° = \sin 30°$)

$\Rightarrow \boxed{W_g = 61.3\ J}$

(b) $W_f = ?$ Magnitude of frictional force $f_k = \mu_k n$

$n = mg\cos\theta \Rightarrow f_k = \mu_k mg \cos\theta$

$\vec{f_k}$ is directed against displacement $\vec{\Delta x}$

$\Rightarrow$ angle between $\vec{f_k}$ and $\vec{\Delta x}$ is $180°$ ($\cos 180° = -1$)

$\Rightarrow W_f = f_k \Delta x (\cos 180°) = -\mu_k mg \cos\theta \cdot \Delta x$

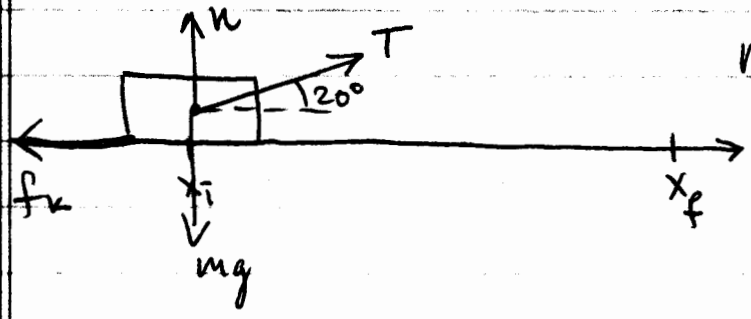
$= -0.436 \cdot 5 \cdot 9.8 \cdot \cos 30° \cdot 2.5 = \boxed{-46.3\ J}$

(c) The normal force $\vec{n}$ is perpendicular to $\vec{\Delta x}$

$\Rightarrow$ angle between them is $90°$.

But $\cos(90°) = 0 \Rightarrow W_n = n \cdot \Delta x \cdot \cos(90°) = \boxed{0}$.

P7



$m = 18.0\ kg$, $\Delta x = x_f - x_i = 20.0\ m$

$\mu_k = 0.500$

$\vec{v} = const \Rightarrow \underline{\vec{a} = 0}$

(a) Use Newton's 2nd law to find tension T

$\otimes$ $T\cos 20° - f_k = 0$ (because $a_x = 0$)

$\oplus$ $T\sin 20° + n - mg = 0$

$\Rightarrow n = mg - T\sin 20°$

From $\otimes$ $f_k = T\cos 20°$

but $f_k = \mu_k n = \mu_k(mg - T\sin 20°)$

$\Rightarrow$ equation for unknown T: $T\cos 20° = \mu_k(mg - T\sin 20°)$

$T(\cos 20° + \mu_k \sin 20°) = \mu_k mg$

$$T = \frac{\mu_k mg}{\cos 20° + \mu_k \sin 20°} = \frac{0.5 \cdot 18 \cdot 9.8}{\cos 20° + 0.5 \sin 20°} = \boxed{79.4\,N}$$

(b) $W_T = ?$ $W_T = T\cos 20° \cdot \Delta x = 79.4 \cdot \cos 20° \cdot 20 = 1492\,J$

$= \boxed{1.49 \cdot 10^3\,J}$

(c) Mechanical energy lost due to friction =

= work done by friction.

$f_k = \mu_k n = \mu_k(mg - T\sin 20°) = 0.5(18 \cdot 9.8 - 79.4\sin 20°)$

$= 74.6\,N$

$W_f = -f_k \Delta x = -74.6\,N \cdot 20\,m = -1492\,J =$

$= \boxed{-1.49 \cdot 10^3\,J}$

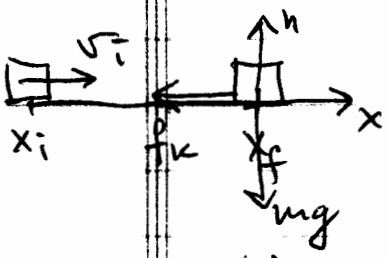
Note: $W_f = -W_T \Rightarrow$ total work $W_{net} = W_f + W_T + \underset{0}{W_n} + \underset{0}{W_g} =$

$= 0$

$\Rightarrow KE_f - KE_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = W_{net} = 0 \Rightarrow \boxed{v_f = v_i}$

P13 $m = 70\,kg$, $v_i = 4.0\,m/s$, $\mu_k = 0.70$
$v_f = 0$

(a) $W_{net} = W_f$ (work by friction) because normal ($\vec{n}$) and gravitational ($m\vec{g}$) forces do no work (displacement is horizontal)



Work - Kinetic Energy Th: $W_{net} = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$

$v_f = 0 \Rightarrow W_{net} = -\frac{1}{2} m v_i^2 = -\frac{1}{2}\, 70\,kg \cdot (4.0\,m/s)^2 = -560\,J$

Energy lost $= \boxed{560\,J}$

(b) $W_{net} = W_f = f_k \cdot \Delta x \cdot \cos(180°) = -f_k \cdot \Delta x = -560\,J$

$f_k = \mu_k n = \mu_k mg = 0.70 \cdot 70\,kg \cdot 9.8\,m/s^2 = 480\,N$

$\Rightarrow (-480\,N)\,\Delta x = -560\,J$

$\Rightarrow \Delta x = \frac{560\,J}{480\,N} = 1.17\,m = \boxed{1.2\,m}$